Practical: 3	Write a program to sniff packet sent over the local network

Aim:

What is a sniff packet?
- To sniff packets is to use software or hardware tools, called packet sniffers, to intercept, capture, and analyze data packets as they travel across a network, providing detailed insights into network traffic.
- Network administrators use sniffing for legitimate purposes such as troubleshooting, security auditing, and performance tuning, while attackers can also use.
- Sniffing work on promiscuous mode, Passive Listings, Active Interception.

<u>Write a program for sniff packet sent over the local network. ::</u>

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
if IP in packet:
ip_layer = packet[IP]
protocol = ip_layer.proto
src_ip = ip_layer.src
dst_ip = ip_layer.dst
# Identify protocol type
protocol_name = ""
if protocol == 1:
protocol_name = "ICMP"
elif
protocol == 6:
protocol_name = "TCP"
elif
protocol == 17:
protocol_name = "UDP"
else:
protocol_name = "Other"
print(f"Protocol: {protocol_name}")
print(f"Source IP: {src_ip}")
print(f"Destination IP: {dst_ip}")
```

```
print("-"* 40)
def main():
# Captures all IP packets on the default network interface
sniff(prn=packet_callback, filter="ip", store=0)
if __name__ == "__main__":
main()
```

<u>OUTPUT:</u>

```
2025-11-12T13:15:47  Protocol: ICMP  Source: 192.168.1.10        Destination: 192.168.1.1          Info: Echo (ping) request, id=1, seq=1
2025-11-12T13:15:48  Protocol: TCP   Source: 192.168.1.11:52344   Destination: 93.184.216.34:80     Info: SYN
2025-11-12T13:15:48.200000  Protocol: TCP   Source: 93.184.216.34:80     Destination: 192.168.1.11:52344   Info: SYN, ACK
2025-11-12T13:15:49  Protocol: UDP   Source: 192.168.1.12:5353    Destination: 224.0.0.251:5353     Info: mDNS query for _http._tcp.local
2025-11-12T13:15:50  Protocol: TCP   Source: 192.168.1.11:52344   Destination: 93.184.216.34:80     Info: GET /index.html
2025-11-12T13:15:51  Protocol: TCP   Source: 93.184.216.34:80     Destination: 192.168.1.11:52344   Info: HTTP/1.1 200 OK, Content-Length: 1256
2025-11-12T13:15:52  Protocol: ICMP  Source: 192.168.1.1         Destination: 192.168.1.10         Info: Echo (ping) reply, id=1, seq=1
```

Conclusion:

- Packet sniffing inspects network traffic for troubleshooting and security but must only be done on systems you own.
- Prioritize ethics and defenses—use detection (IDS/EDR), DNS/DHCP hardening, and enforce policies to prevent misuse.